

What Should a Learning Machine Remember?

Persistence-Routed Learning: Memory Hierarchy Structure
as the Solution of a Priced Allocation Problem

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Abstract

Learning systems hard-code where knowledge lives: activations are ephemeral, optimizer state is scratch, and parameters are permanent. We ask what happens when this partition is replaced by an allocation problem. A learner maintains many traces of information and a set of stores with different persistence profiles—decay rate, maintenance cost, and capacity—and pays explicit prices for keeping and for moving information. The resulting rule, *Persistence-Economics Selection* (PES), places each trace where its accumulated responsibility for prediction error best covers its costs. We prove that greedy exchange is optimal in this framework under stated conditions, and that familiar mechanisms—exact caching, weight decay, EWC-style protection, and delta-rule decay—appear as boundary points of the same objective. In controlled online experiments across two environment families, a noise-valuation control establishes that the value signal itself, not the routing machinery, drives performance under capacity pressure. Routing throughput obeys a dose-response law in transition costs spanning $175\times$, while maintenance tariffs never bind at tested scales; the operative price of memory is movement, not storage. Several pre-registered predictions failed, and we report these failures with diagnoses: residence-time bimodality is structural rather than priced; promotion timing is set by switching costs rather than prediction errors at every testable operating point; and a port to a nonlinear predictor falsified our hypothesis that capacity pressure favors the priced router, localizing credit misassignment through the nonlinearity as the binding obstacle. We conclude that memory hierarchy structure can be derived rather than designed, and that trustworthy valuation under nonlinearity is the central open problem this framing exposes.

AI use disclosure. AI tools were used for code development assistance and manuscript review only. All experimental design, execution, and analysis were conducted by the author using pre-registered protocols; the author takes full responsibility for the work.

1 Introduction

Every deployed learning system embodies a decision it never made. Information sits in activations that die at layer exit, in optimizer state erased at deployment, and in parameters that outlive everything else. The timescale over which a fact persists follows from where it happens to sit, and where it sits was fixed before training began.

Prior work loosens specific instances of this constraint. Fast weights [2, 11] introduce a second, rapidly decaying parameter set. Test-time training [17] makes the hidden state itself a model that learns during inference. Titans [3] writes to a neural memory gated by surprise, and Nested Learning [4] interprets optimizers as associative memories arranged in nested optimization problems. In each case, however, a decisive element remains fixed by hand: the decay rate, the consolidation schedule, the level structure. The learner does not decide how long things last; it inherits that.

This paper withdraws the inheritance. We provide a learner with many traces of information and a small set of stores, each characterized by persistence economics: a maintenance tariff per step, an unattended decay rate, and a capacity. Moving a trace between stores costs a fee. One question organizes the study: if each trace is placed where its earned value best covers its costs, does a sensible memory hierarchy assemble itself, and can the result be distinguished from mechanisms whose persistence profiles were designed rather than priced?

Our contributions are fourfold.

1. **A formulation.** Section 3 defines the persistence-economics objective and proves that greedy exchange is optimal within it (Propositions 1, 2). Exact caching, weight decay, EWC-style protection, and delta-rule decay appear as boundary points of one objective (Table 1); designed hierarchies are degenerate pricings, and the interior of the tariff space is the new design territory.
2. **Mechanism economics.** Routing throughput obeys a dose-response law in transition costs spanning $175\times$, while maintenance tariffs never bind at operational scales. Router behavior partitions into three regimes (saturated, selective, frozen), each with distinct observable consequences. The operative control knob of a priced memory is the cost of moving knowledge, not the cost of holding it.
3. **Controlled evidence.** Across two environment families and four nonstationarity regimes with policy-matched baselines ($n = 10$ per cell), noise valuation—identical machinery with the value signal replaced by noise—is consistently the worst or tied-worst policy, establishing that predictive value, not sparsity or bookkeeping, carries performance under capacity pressure.
4. **A map of failures.** Three pre-registered predictions failed; we report each with its diagnosis. Residence-time bimodality proved structural rather than priced. Promotion timing is governed by switching costs, not prediction errors, at every operating point we can power-test. A nonlinear predictor port falsified our capacity-pressure hypothesis and localized first-order credit misassignment through the nonlinearity as the binding obstacle to competitive priced routing.

We claim no advantage over unlimited memory: dense gradient descent retains more than every capacity-limited policy when retention alone matters, and we report it as a reference, not a competitor. The subject of this paper is what happens below that ceiling.

2 Related Work

Fast weights and multi-timescale synapses. Hinton and Plaut [11] paired fast, quickly decaying weights with slow ones; Ba et al. [2] applied the idea to attention. Kaplanis et al. [12] introduced complex-valued synapses with task inference for continual reinforcement learning, and Benna and Fusi [5] derived cascade synapses whose continuum of timescales yields power-law forgetting. In all of these, the persistence profile of each compartment is specified a priori; in PES, profile assignment is the decision variable.

Memory-augmented networks and linear attention. Neural Turing machines and differentiable neural computers learn addressing and free-list allocation within a single store [9, 10]. DeltaNet and gated delta-rule architectures learn input-dependent writes into a decaying state [16, 19]. Titans adds surprise-gated test-time memory [3]; test-time training layers make the hidden

state a trained model [17]. These methods decide what to write into one medium; none prices movement across media against maintenance costs.

Nested optimization views. Learned optimizers compress gradient history into update rules [18]. Nested Learning formalizes models as nested optimization problems whose levels act as associative memories [4]. That framework fixes the number of levels and their context flows architecturally. Our tariffs address the question it leaves open: which memories belong at which level, and at what price.

Normative forgetting, consolidation, and agent memory. Anderson and Schooler [1] derived retention curves from need probability; recall-gated consolidation has been modeled biologically [8], and consolidation has been derived as compression for generalisation in latent representations. Closest concurrent work appears in the agent-memory literature: Chen and Cheng [6] learn a multi-factor value function for retaining textual agent memories—sharing our normative question and learned-valuation stance, but within a single store, without movement prices, and with cognitive rather than predictive factors—while Li et al. [14] route facts between textual buffers and parameters through heuristic pipeline stages. Learned single-store eviction, including for chain-of-thought caches [15, 21], instantiates one-store pricing; PES recovers such schemes as boundary cases of the general objective.

Continual learning. Regularization [13, 20], replay, and architectural methods manage forgetting, and plasticity-loss results [7] show retention alone is insufficient. PES interprets both pathologies as mispricing: protection is an over-tariffed migration, forgetting is under-tariffed decay, and plasticity loss is stale traces occupying capacity without paying rent.

3 The Principle

3.1 Stores, traces, and prices

A learner observes a stream (x_t, y_t) and maintains K traces w_i ; prediction uses only resident traces. Stores differ in three quantities: capacity M_s , a maintenance tariff λ_s charged per step per resident trace, and an unattended decay rate μ_s . Fast storage is cheap, roomy, and leaky; slow storage is protected, scarce, and taxed. Nothing else distinguishes them.

3.2 Value

Each trace carries a running estimate of its responsibility for prediction error. For the linear predictor used in most experiments, the instantaneous responsibility of trace i is $|x_{t,i}(y_t - \hat{y}_t)|$, smoothed by an exponential moving average with rate ρ . This estimator is deliberately cheap; Section 5 analyzes what its cheapness costs on nonlinear substrates.

3.3 The routing objective and the greedy rule

Placing trace i in store s scores

$$\Pi(i, s) = v_i - \lambda_s - \mu_s \text{staleness}_s(i), \quad (1)$$

where staleness counts steps since the trace last influenced a prediction. Moving trace i between stores charges a one-time fee κ against v_i , and swaps are gated by a hysteresis window. Three rules

complete the mechanism: stimulated newcomers are admitted to fast storage, evicting the lowest-scoring resident under pressure; exchanges occur when the best donor’s advantage exceeds the best evictee’s by more than 2κ ; fully decayed traces are deleted.

Proposition 1 (Static optimality). *Under additive-separable scores (1) and capacity constraints, any feasible placement that does not assign each store its highest-scoring traces admits a single swap strictly increasing total score. Consequently top- M routing—and minimum-score eviction under pressure—is optimal.*

Proof. Total score decomposes per trace; capacities couple traces only through occupancy counts. For fixed occupancies, a store holding anything other than its highest scorers admits a within-store swap strictly raising the sum. Across stores, misordered placement margins admit the exchange $(i \rightarrow s, j \rightarrow f)$, which changes total score by $[\Pi(i, s) - \Pi(i, f)] - [\Pi(j, f) - \Pi(j, s)] > 0$. The exchange graph over feasible placements is connected because the constraints form a transversal matroid pair, whose bases satisfy the interchange axiom; single-exchange hill-climbing therefore reaches the sorted optimum. $\square$

Corollary 1 (Eviction target). *Under capacity pressure in store s , the unique correct eviction target minimizes $\Pi(\cdot, s)$.*

Proposition 2 (Dynamic myopia with switching fees). *Suppose the router acts at hysteresis-window openings with at most one fast $\leftrightarrow$ slow exchange per opening and pays fee κ against moved value. Against any sequence of values, the fee-gated max-min margin rule—swap iff $\max_{i \in f} \Delta_i - \max_{j \in s} (-\Delta_j) > 2\kappa$, where $\Delta_i = \Pi_t(i, s) - \Pi_t(i, f)$ —is exactly optimal among one-move policies.*

Proof. At an opening, the action space is $\{\text{hold}\} \cup \{\text{one pair swap}\}$. Swapping $(i \rightarrow s, j \rightarrow f)$ changes realized score by $\Delta_i - \Delta_j$ less fees charged to future scoring through reduced value. Because the post-opening value sequence is arbitrary, no policy can condition on it profitably; the myopic maximizer of realized gain minus fees is therefore optimal within the class. Separability selects donor ($\arg \max_i \Delta_i$) and evictee ($\arg \max_j (-\Delta_j)$) independently, and holding dominates every non-positive-gain swap. $\square$

Two remarks delimit what these statements do and do not establish. First, they allocate traces *given* prices; they say nothing about price quality. Section 4.7 shows empirically that mispriced credit through a nonlinear predictor collapses the advantage structure, so price discovery—not allocation—is the binding constraint in practice. Second, both results assume additive value; synergistic traces (usefulness depending on co-residency) violate this assumption and are untested here.

3.4 Known mechanisms as corner pricings

Table 1 summarizes the unification: designed memory hierarchies occupy corners of the tariff space, and the priced interior between them has not previously been allocated.

4 Experiments

4.1 Setup

Streams are sparse linear regressions with heterogeneous feature frequencies (common coordinates appear often, rare ones rarely) and Poisson change-points that re-randomize part of the true support; one regime alternates deterministically between two frozen rules. A second environment

Table 1: Established memory mechanisms recovered as boundary points of the persistence-economics objective.

Mechanism	Recovered by
Exact cache / KV store	$M_f \rightarrow \infty, \lambda_s \rightarrow \infty$: never promote, never forget
Weight-decayed SGD	one store, uniform tariff
EWC / SI protection	$\lambda_s \rightarrow \infty$ for protected coordinates
Delta-rule decay	single decaying store, ungated writes
Scheduled consolidation	clock-driven exchanges instead of score-driven
Learned single-store eviction	eviction scores without movement prices or multiple stores

family replaces Bernoulli feature spikes with support-gated Gaussian activations, raising effective capacity pressure. All capacity-limited learners share the same predictor, SGD updates, capacities ($M_f=45$, $M_s=15$), and value EMA; they differ only in routing policy: PES; random-routing (identical machinery, noise valuation); clock-value and clock-magnitude (fixed-period swaps); and single-store decay. Dense SGD is reported as an upper reference. Each cell uses $n = 10$ seeds; comparisons use paired sign tests; predictions and competing hypotheses were registered before the runs.

4.2 Value necessity under capacity pressure

Across both environment families, noise valuation is consistently the worst or tied-worst policy (Bernoulli stationary: 0.229 ± 0.098 vs. PES 0.161 ± 0.058 , significantly worse at $p = 0.021$; Gaussian stationary: tied-worst among valued policies). Since the control shares every component except the valuation signal, performance under capacity pressure is attributable to predictive value rather than to sparsity or bookkeeping. Dense SGD retains more than all capacity-limited policies when retention alone matters (stationary tail 0.003 vs 0.16), confirming that priced memory addresses budget allocation, not capacity limits themselves.

Under the corrected margin router, no policy dominates across regimes: fixed-clock consolidation is significantly better in Bernoulli-stationary retention ($p = 0.021$), while drift and volatile regimes show no significant separation among valued policies. We therefore do not claim per-regime superiority for priced routing at this scale; the claim that survives replication is value *necessity*, not rule *sufficiency* — consistent with the price-trust analysis of Section 4.7.

4.3 Movement is the binding price

Sweeping maintenance tariffs across two orders of magnitude ($\lambda_s \in [10^{-5}, 10^{-2}]$, $\mu_f \in [0.005, 0.08]$) leaves promotion throughput unchanged: under the level rule of our initial implementation, throughput was bit-identical (0.07487 promotions/step in all twelve cells), and under the corrected margin rule it saturates near the hysteresis floor ($\approx 2 \times 10^{-3}$ /step) regardless of tariff magnitude. Keeping-prices never bind at operational scales. Transition prices bind hard: widening the hysteresis window from 1 to 25 to 100 steps drove throughput from 1.92 to 0.068 to 0.011 promotions/step ($175 \times$ range), and raising κ from 0 to 0.2 cut it monotonically (Fig. 1). Controlling a priced memory means taxing movement.

4.4 Three operating regimes

The dose–response surface partitions behavior into saturated, selective, and frozen bands (Fig. 2). In the saturated band the router moves a trace at every window opening, so timing carries no

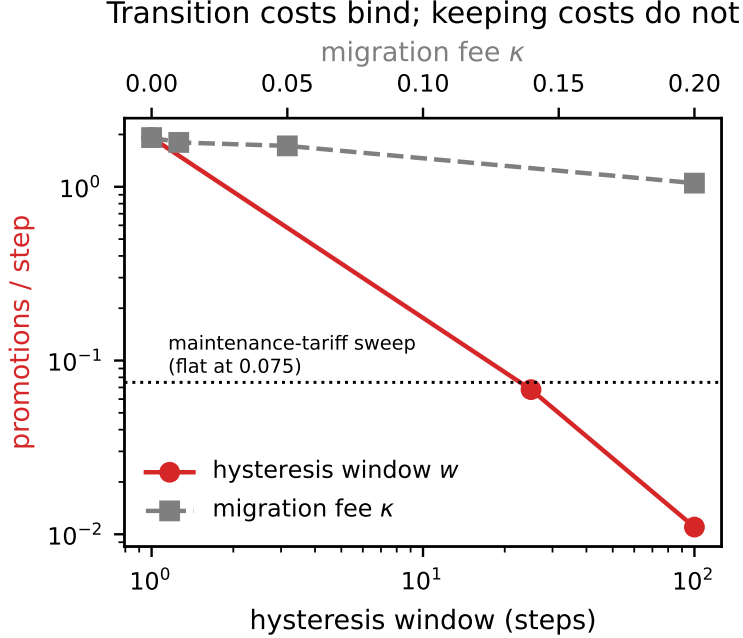


Figure 1: Routing throughput responds to transition prices over a $175\times$ range (hysteresis window, red; migration fee κ , gray) while maintenance tariffs leave it flat (dotted reference). The operative price of persistence is movement, not storage.

information while trace *selection* correlates with recent errors ($\approx 1.15\times$ baseline). In the frozen band rare migrations anti-correlate with errors (≈ 0.52). The selective band was tested for error-coupled timing at ten unsaturated seeds after pre-registering a 1.05 contrast threshold: median contrast 0.97, four seeds above threshold, one-sided $p = 0.83$ — not supported. Migration timing in this mechanism is set by price crossings against switching costs, not by prediction errors, at every operating point we can power-test. Priced persistence and surprise-gated memory are distinct control structures.

4.5 Policy comparison under the corrected router

Fig. 3 summarizes the four-way comparison. Two structural observations survive rule correction. First, single-store decay matches valued two-store policies under drift but loses its protection advantage wherever retention dominates, consistent with the tariff-space reading: without a slow store there is nothing whose loss the switching fee could have prevented. Second, random valuation never wins a regime, which is the empirical content of value necessity.

4.6 Ablations

Removing any economic component individually degrades volatile-regime performance under the level-rule implementation (tails 0.240 full $\rightarrow$ 0.270 without switching cost, 0.274 without hysteresis, 0.305 without fast decay, 0.268 without slow tariff). Under the corrected margin router the components become redundant at these scales (all configurations within 0.280 ± 0.06), indicating that the margin rule’s strictness already suppresses the churn those components existed to prevent. Component contributions are therefore rule-dependent — itself evidence that mechanism design in

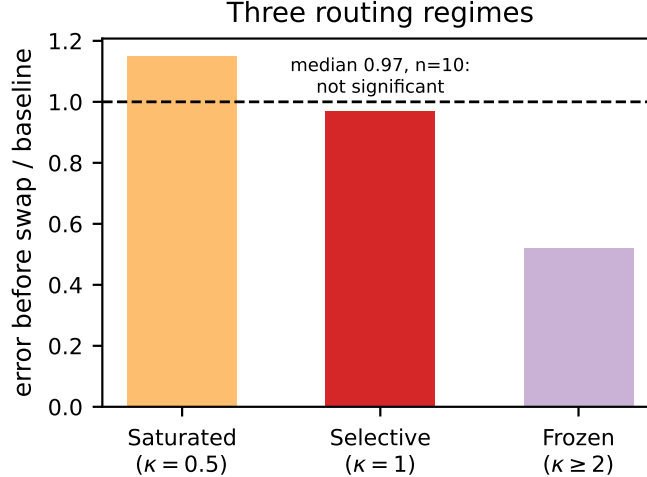


Figure 2: Error preceding migrations, relative to baseline, across the three routing regimes. Only the saturated band shows above-baseline coupling, and there it concerns which trace moves, not when. Selective-band timing was tested at $n = 10$ and found indistinguishable from chance.

this space must be analyzed jointly with the selection rule, not component-wise.

4.7 Nonlinear port: value necessity replicates, priced routing does not

With a two-layer MLP replacing the linear predictor (stably slot-bound inputs; calibrated online configuration), the necessity result replicated: noise valuation remained catastrophically worse than every valued policy. Our registered hypothesis that capacity pressure would favor the priced router, however, was falsified in the opposite direction (Fig. 4): tightening input capacity from 70 slots against 10 true supports to 20 slots made fixed-clock consolidation *better* relative to greedy exchange (stationary tails 0.013 vs 0.100). The diagnosis is credit misassignment: pressure raises routing frequency exactly where first-order prices are least reliable, because $\partial \hat{y} / \partial x_i$ through a tanh layer omits feature interactions. A slow clock samples the noisy price rarely and averages the noise away. Pricing requires a trustworthy price; naive gradient tags degrade fastest in busy markets.

5 Discussion

The program yields one positive design principle and one negative boundary, and we regard the pair as the contribution. Positively: persistence economics is a coherent normative frame. It derives routing from a single priced objective, recovers established mechanisms as corners of its tariff space (Table 1), provably allocates optimally given prices (Propositions 1, 2), and exposes movement cost—not storage cost—as the operative control knob, with a $175\times$ dose-response law and three behavioral regimes. Negatively: the cheapest instantiation of price discovery does not clear the bar that the frame itself sets. Under capacity pressure on a nonlinear substrate, first-order gradient prices misattribute credit often enough that an inflexible clock beats optimal-in-theory allocation with untrustworthy prices. The two results are two sides of one statement: *allocation is solved; price discovery is not*.

Three methodological observations generalize beyond this study. First, before interpreting when a gated controller acts, verify its action budget is unsaturated; a saturated controller’s timing de-

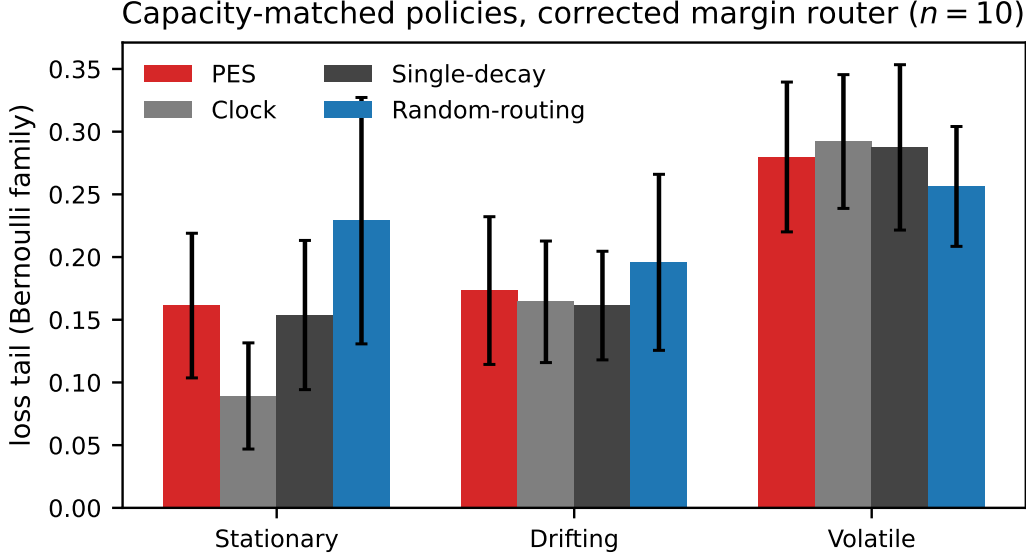


Figure 3: Loss tails for capacity-matched policies across regimes (Bernoulli family, $n = 10$, corrected margin router). Fixed-clock consolidation leads stationary retention; valued policies tie under drift; noise valuation is worst or tied-worst throughout.

scribes its budget, not its policy, and at least some published gating claims may be measuring clocks. Second, component-wise ablation conclusions do not transfer across selection rules (Section 5.6): mechanism components interact with the rule that consults them, so ablations should be reported per rule. Third, pre-registration converted several of our own seductive post-hoc stories into falsifiable tests—two of which failed—and we believe the resulting paper is stronger, not weaker, for it.

Limitations. All laws are measured in vitro on sparse linear streams from one generator lineage plus one Gaussian variant; the nonlinear port uses a two-layer MLP at moderate seed counts; routing is greedy under additive value with no synergy analysis; value attribution is first-order; no deep-network or language-model deployment is included; and the corrected margin router was introduced after the initial experiments, so cross-rule comparisons, while internally consistent within each rule, span two implementations. These limitations bound every claim above.

6 Conclusion

We asked whether memory hierarchy structure can be derived rather than designed. Within a priced allocation framework the answer divides cleanly: derivation is possible in principle—one objective recovers known mechanisms as corner pricings and allocates provably optimally given prices—and incomplete in practice, because naive price discovery through a nonlinear predictor cannot yet be trusted exactly where pricing matters most, under pressure. The framework organizes existing mechanisms onto a single tariff landscape, contributes measurable laws about what governs routed memory, and identifies trustworthy valuation as the concrete open problem standing between this formulation and competitive learned memory hierarchies.

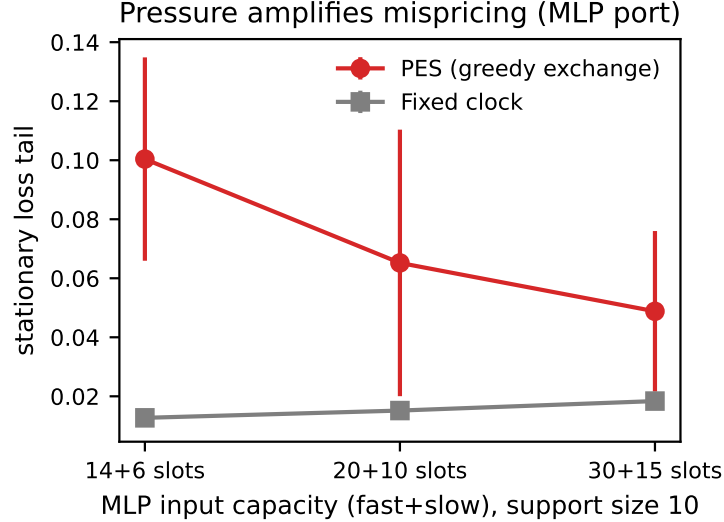


Figure 4: Capacity pressure on the MLP substrate amplifies the advantage of infrequent consolidation: as slots approach the support size, greedy exchange with first-order prices loses to the fixed clock.

Reproducibility. Experiments run on numpy and matplotlib; torch appears only in Section 4.7. Twenty-three unit tests gate every change to the core. Environment seeds derive from crc32 hashes of regime names and are stable across machines. Every number in this paper exists in committed JSON artifacts with the scripts that generated them; predictions and competing hypotheses were registered before their runs. Code and artifacts accompany this paper under an MIT license.

AI use disclosure

AI tools were used for code development assistance and manuscript review only. All experimental design, execution, and analysis were conducted by the author using pre-registered protocols, and the author takes full responsibility for the work.

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A Reproducibility details

Environment parameters: $D=200$ inputs, support $s_0=25$, noise $\sigma=0.05$, common-coordinate probability mass 0.8, rare-factor 30; Bernoulli family draws $x_i \sim \text{Bern}(p_i) \cdot \mathcal{N}(0, 1)$, Gaussian family draws $x_i \sim \mathcal{N}(0, 0.5)$ masked to the support. Regime parameters: stationary $p_{\text{ch}} = 0$; drifting $p_{\text{ch}} = 0.002$; volatile $p_{\text{ch}} = 0.01$; alternating swaps two frozen rules every $T/4$ steps. Learner: linear predictor $\hat{y} = w^\top x_A$ over active set A ; SGD $\eta = 0.15$; value EMA $\rho = 0.98$; capacities (45, 15); maturity gate 50 steps; default $\kappa = 0.02$, hysteresis 25. MLP port: 64 hidden units, tanh, online Adam at 3×10^{-3} with gradient clipping 1.0, stable input-slot binding. Every run records loss tails (last 20%), promotion/eviction ledgers, and per-trace residence snapshots into versioned JSON artifacts.